DarkSage

Database Design

Version 0.5.0 — Draft

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Wisdom Over Noise

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Revision History

Version	Date	Author	Summary
0.5.0	2026-07-25	TheSinnerMan / Keeper	Independent-audit blocker-repair (Blockers 2, 3): DS-DB-032/033 constraints extended with DS-JRN-007 retention/deletion behavior; DS-DB-028 constraints extended with DS-SCA-029's provenance/tamper-detection/stale-state security requirements. No entity's Key Fields or Key Relationships changed; constraints/testing extended only.
0.4.0	2026-07-25	TheSinnerMan / Keeper	Independent-audit repair (H1, H3): added `DS-DB-028` (TradeIntelligencePackage), `DS-DB-029` (ResearchSource/ResearchEvidence), `DS-DB-030` (CatalystEvent), `DS-DB-031` (TradingThesis/ThesisRevision), `DS-DB-032` (JournalEntry), `DS-DB-033` (DailyReview/WeeklyReview) as fully-specified entities (Key Fields/Relationships/Constraints/Testing), replacing the prior unnumbered prose-only mention in this section. Renumbered the Founder Vision Completion section from the misnumbered "## 22." (skipping §§20–21) to the correct next-available "## 20." No prior entity's content changed.
0.3.0	2026-07-25	TheSinnerMan / Keeper	Founder Vision Completion amendment and cross-volume traceability, including Discord notification boundaries where applicable.
0.1.0	2026-07-24	TheSinnerMan	First controlled draft. Defines entity/schema design for the shared models `ARCHITECTURE.md` §6 names, on the SQLite-first strategy DS-ARC-016 already establishes, cross-referenced to DS-002's requirement families.

Version	Date	Author	Summary
0.2.0	2026-07-24	TheSinnerMan	Repair pass addressing independent audit findings DS-005-A01 through A05. **A01:** added Transaction entity (DS-DB-025), Position and DS-DB-023 updated to reference it. **A02:** added Alert (DS-DB-026) and Notification (DS-DB-027) entities as new §13, sections renumbered 13→14 through 18→19 accordingly. **A03:** reclassified StrategyProfile and TradeDecision to Committed/MVP (Phase-1 base model, per ROADMAP.md/DS-ARC-005), reclassified BrokerState Future/Exploratory → Planned, made ScanConfiguration.profile_id nullable so it is independently satisfiable without the Planned UserProfile feature. **A04:** corrected Candle's foreign-key target (DS-DB-013 → DS-DB-014, SecurityIdentity); added signal_id to Signal and regime_id to MarketRegime as their primary identifiers, matching the foreign keys that already referenced them. **A05:** split StrategyProfile and TradeDecision into a Committed Phase-1 core and a Planned later-phase extension (Phase 2 for StrategyProfile's configuration/risk fields; Phase 6/7 for TradeDecision's AI-provider/pipeline fields), so neither Committed base model requires a Planned capability to be valid.
0.2.1	2026-07-24	TheSinnerMan	Consolidated cleanup pass: corrected DS-DB-004's stale Alert reference (Alert was already added as DS-DB-026 in the 0.2.0 A02 repair; the Key Relationships text still said "not yet in this document's entity set") to cite DS-DB-026 directly. Corrected DS-DB-027's channel field, which mislabeled DS-ALT-002 (in-app notification delivery) as "Committed" although DS-ALT-002 is formally Planned — both in-app (DS-ALT-002) and external (DS-ALT-003) channels are now correctly stated as Planned. No entity, classification, or relationship was otherwise changed.

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1. Purpose

DS-005 defines DarkSage's data entities: their purpose, key fields, relationships, and integrity constraints. It builds directly on `ARCHITECTURE.md` §6 (Shared Models) and DS-ARC-005/DS-ARC-016 (which establish *that* shared models and SQLite-first storage exist); DS-005 defines *what those models actually contain*. This is a schema-design document, not a full DDL implementation artifact — exact column types, indexes, and migration tooling are left to implementation, while entity shape, required fields, relationships, and invariants are normative here.

2. Scope

Covers entity design for: core market data (Candle, Quote, MarketRegime), signals and strategy (Signal, StrategyProfile, TradeDecision, BacktestResult), portfolio and execution (Position, Transaction, Portfolio, Order, BrokerState, RiskState), charting (ChartAnnotation), product/platform state (UserProfile, Preferences, Watchlist, WorkspaceLayout, SecurityIdentity), security/operations (IntegrationCredential, AuditLogEntry), scanning (ScanConfiguration, ScanResult), and alerts/notifications (Alert, Notification).

Does not cover: exact SQL DDL/migration tooling (implementation detail), API request/response shapes (DS-006), or UI data-binding (DS-007).

3. Audience

Backend/database contributors, independent auditors.

4. Definitions

See DS-001 §24, DS-002 §4, DS-004 §4.

5. Database Strategy Recap

Per DS-ARC-016: SQLite is the initial database; PostgreSQL/TimescaleDB migration is a later, not-yet-justified decision (Phase 12). All entity designs in this document assume SQLite's type system (dynamic typing, no native array/JSON column enforcement beyond TEXT-stored JSON) as the baseline implementation target, without precluding a future migration.

6. Core Market Data Entities

DS-DB-001 — Candle

Release Classification: Committed / MVP | **Governing Source:** `ARCHITECTURE.md` §6; `ROADMAP.md` Phase 1 ("Candle model"); DS-MKT-001, DS-MKT-002

Purpose: Store OHLCV market data at a given symbol/timeframe/timestamp — the base unit historical and real-time price analysis builds on.

Key Fields: symbol_id (FK → SecurityIdentity, DS-DB-014), timeframe, timestamp (UTC), open, high, low, close, volume, source_provider, ingested_at, is_adjusted (bool, corporate-action adjustment flag).

Key Relationships: Many Candles per SecurityIdentity; consumed by DS-ARC-007 (Indicator Engine), DS-ARC-008 (Chart Adapter), DS-ARC-010 (Backtest Engine).

Constraints/Invariants: Unique on (symbol_id, timeframe, timestamp); immutable once stored except through an explicit, logged correction (DS-MKT-002); `is_adjusted` distinguishes split/dividend-adjusted series from raw, consistent with DS-MKT-006's corporate-action handling once built.

Testing: Uniqueness-constraint test; correction-audit-trail test.

DS-DB-002 — Quote

Release Classification: Committed / MVP | **Governing Source:** `ARCHITECTURE.md` §6; `ROADMAP.md` Phase 1; DS-MKT-003, DS-PRD-008

Purpose: Store the current/last-known price point for a symbol, with the state metadata DS-PRD-008 requires.

Key Fields: symbol_id (FK), price, bid, ask, size, timestamp, data_state (enum: current/delayed/stale/historical/simulated), delay_seconds (nullable), source_provider.

Key Relationships: One current Quote per SecurityIdentity per provider; referenced by Position (DS-DB-009) for mark-to-market.

Constraints/Invariants: `data_state` and `delay_seconds` are never null when `data_state != current`; a Quote record is a snapshot, not mutated in place — a new record supersedes rather than overwrites (supports DS-PRD-008's auditability of "what changed").

Testing: State-transition test (current → delayed → stale) per DS-MKT-004's threshold logic.

DS-DB-003 — MarketRegime

Release Classification: Planned | **Governing Source:** `ARCHITECTURE.md` §6; `PROJECT_SPEC.md` §13; `ROADMAP.md` Phase 3

Purpose: Store classified market-environment state (bull/bear/sideways, volatility regime, etc.) used to segment strategy performance (`PROJECT_SPEC.md` §7/§13).

Key Fields: regime_id (primary identifier), regime_type, effective_date, confidence, classification_method_version.

Key Relationships: Referenced by BacktestResult (DS-DB-006) and StrategyProfile performance segmentation for regime-based analysis.

Constraints/Invariants: `classification_method_version` is recorded so a later change to the classification methodology doesn't silently reinterpret historical regime labels.

Testing: Not yet applicable at Planned classification; deferred to Phase 3 authoring.

7. Signal, Strategy, and Backtesting Entities

DS-DB-004 — Signal

Release Classification: Committed / MVP | **Governing Source:** `ARCHITECTURE.md` §6; `ROADMAP.md` Phase 1 ("Signal model"); `PROJECT_SPEC.md` §16; DS-SCN-002/003

Purpose: Represent a single scan/strategy output: a candidate opportunity with its supporting evidence and score.

Key Fields: signal_id (primary identifier), symbol_id (FK), strategy_id (FK, nullable for pure-scanner signals), direction, entry/stop/targets (nullable until a strategy assigns them), confidence, quantitative_score, technical_score, fundamental_score, sentiment_score, detected_patterns (JSON), indicators_snapshot (JSON), reasoning (text), grade (A+/A/B/C/D per `PROJECT_SPEC.md` §16), generated_at, expires_at (nullable).

Key Relationships: Many Signals per SecurityIdentity; optionally linked to a StrategyProfile (DS-DB-005); optionally referenced as an Alert (DS-DB-026, Planned) condition target via `condition_definition`, per DS-DB-026's own Key Relationships.

Constraints/Invariants: `grade` is derived from measurable inputs (`TRADING_RULES.md` "Signal Grades"), never assigned by unstructured AI judgment alone (DS-PRD-004 analog for scoring); a rejected signal retains its rejection reason (`PROJECT_SPEC.md` §17, Why-Trade/Why-Not-Trade), not deleted.

Testing: Grade-derivation determinism test; rejection-reason completeness test.

DS-DB-005 — StrategyProfile

Release Classification: Committed / MVP (Phase-1 base model only — see field split below; full strategy-construction functionality remains Planned) | **Governing Source (corrected in the DS-005-A03 repair):** `ROADMAP.md` Phase 1 ("StrategyProfile model") places the base model in the Committed Phase-1 subset; `ARCHITECTURE.md` §10; DS-ARC-005 (Committed, narrowed Phase-1 model list).

Purpose: Represent a versioned, auditable strategy definition — as a Phase-1 base record shape now (referenced by Signal's optional `strategy_id`), with full strategy-construction richness arriving in Phase 2.

Key Fields (Phase-1 core, Committed): strategy_id (primary identifier), name, version, status (Experimental/Watch/Active/Reduced/Suspended), created_at, superseded_by (nullable, FK → StrategyProfile, self-referential version chain).

Key Fields (Phase-2 extension, Planned — nullable/absent until DS-STR-001 exists): supported_timeframes (JSON list), supported_instruments (JSON list), configuration (JSON), risk_assumptions (JSON).

Key Relationships: One-to-many with Signal (DS-DB-004), TradeDecision (DS-DB-007), BacktestResult (DS-DB-006).

Constraints/Invariants: A logic change creates a new version row (`superseded_by` chain), never an in-place mutation of a prior version's `configuration` (`AGENTS.md` "No Silent Strategy Changes"); this invariant applies once the Phase-2 `configuration` field is populated and is not required for the Phase-1 core to be valid on its own.

Testing: Phase-1 core: identity/versioning-chain integrity test. Phase-2 extension: immutability-of-prior-version `configuration` test (once DS-STR-001 exists).

DS-DB-006 — BacktestResult

Release Classification: Planned | **Governing Source:** `ARCHITECTURE.md` §6, §13; `ROADMAP.md` Phase 2; DS-BKT-001/002/003/004

Purpose: Store a reproducible backtest run's configuration and output.

Key Fields: backtest_id, strategy_id (FK, version-pinned), date_range_start/end, cost_assumptions (JSON — DS-BKT-002), data_snapshot_reference, results (JSON: trade_count, win_rate, expectancy, profit_factor, max_drawdown, sharpe, sortino, sample_size_confidence), regime_id (FK, nullable), executed_at.

Key Relationships: Many-to-one with StrategyProfile (version-pinned, not just strategy_id, so a later strategy edit doesn't retroactively reinterpret a past result); many-to-one with MarketRegime for regime-segmented analysis.

Constraints/Invariants: Immutable once written (DS-BKT-001's reproducibility requirement); `strategy_id` pins the exact version used, not a mutable reference to "current" strategy state.

Testing: Reproducibility regression test (same strategy version + date range + cost assumptions → identical `results`).

DS-DB-007 — TradeDecision

Release Classification: Committed / MVP (Phase-1 core only — see field split below; pipeline/AI extension fields remain Planned) | **Governing Source (corrected in the DS-005-A03/A05 repair):** `ROADMAP.md` Phase 1 ("TradeDecision model") places the base model in the Committed Phase-1 subset; DS-ARC-005 (Committed, narrowed Phase-1 model list). Full pipeline integration is `ARCHITECTURE.md` §14 / `ROADMAP.md` Phase 7 (DS-ARC-011, Planned); AI provider involvement is Phase 6 (DS-ARC-013, Planned).

Purpose: Represent a Trade Proposal at a Phase-1-safe base level — an idea/candidate a Signal or Strategy could act on — without requiring the Phase-6 AI provider layer or the Phase-7 `TradeValidationPipeline` to exist for the record to be valid. Once those capabilities exist, the same table extends to become the full pipeline audit record (DS-OPS-002's backbone).

Key Fields (Phase-1 core, Committed): decision_id (primary identifier), signal_id (FK), strategy_id (FK, nullable), proposed_size, proposed_direction, final_disposition (Phase-1-safe enum: `proposed` / `rejected` only — `validated`/`executed` are added as valid values once the Phase-7 pipeline exists, see extension below), created_at.

Key Fields (Phase-6/Phase-7 extension, Planned — nullable/absent until those capabilities exist): pipeline_stage_results (JSON — one entry per `TradeValidationPipeline` stage, populated only once DS-ARC-011 is implemented), ai_provider_used (nullable, populated only once an AI provider, DS-ARC-013, exists), ai_output_raw (nullable, text, for audit, same gating). `final_disposition` gains `validated`/`executed` as valid values only once the Phase-7 pipeline exists.

Key Relationships: One-to-one with an eventual Order (DS-DB-010) if `final_disposition = executed` (Phase 7+); many-to-one with Signal and StrategyProfile.

Constraints/Invariants: The Phase-1 core (decision_id, signal_id, strategy_id, proposed_size/direction, final_disposition □ {proposed, rejected}, created_at) is independently valid and complete without any Phase-6/7 field populated. Once populated, `pipeline_stage_results` records every stage's outcome even on early rejection — a rejected proposal still shows which stage rejected it and why (`TRADING_RULES.md` Why-Trade/Why-Not-Trade); this table is the audit-trail backbone for DS-OPS-002 once that extension is in use.

Testing: Phase-1 core: record-validity test with no extension fields populated. Phase-6/7 extension (once applicable): pipeline-stage-completeness audit test; audit-reconstruction test (DS-OPS-002).

8. Portfolio and Execution Entities

DS-DB-008 — Portfolio

Release Classification: Planned | **Governing Source:** `ARCHITECTURE.md` §6, §19; `ROADMAP.md` Phase 4; DS-PRT-004

Purpose: Represent a named collection of positions (a paper account, or in future a live account) with aggregate metrics.

Key Fields: portfolio_id, name, account_type (paper/live), cash_balance, created_at.

Key Relationships: One-to-many with Position (DS-DB-009); referenced by RiskState (DS-DB-011) for portfolio-level risk budget.

Constraints/Invariants: `account_type = live` requires the Stage 4 / Phase 13 prerequisites (DS-ARC-018) to be satisfied before any Order (DS-DB-010) may reference it with a non-paper Broker Adapter.

Testing: Not yet applicable at Planned classification.

DS-DB-009 — Position

Release Classification: Planned | **Governing Source:** `ARCHITECTURE.md` §6; `ROADMAP.md` Phase 2/4/7; DS-PRT-001

Purpose: Represent a held quantity of a security within a Portfolio, derived from Transaction history.

Key Fields: position_id, portfolio_id (FK), symbol_id (FK), quantity, cost_basis, opened_at, closed_at (nullable).

Key Relationships: Derived from the append-only Transaction ledger (DS-DB-025) — a Position is computed by aggregating a symbol's Transactions within a Portfolio, not stored as independently-entered data.

Constraints/Invariants: Quantity/cost_basis are computed deterministically from the transaction log (DS-PRD-004); a Position reduced to zero is closed (`closed_at` set), not deleted.

Testing: Position-derivation regression test against fixture transaction sequences.

DS-DB-025 — Transaction

Release Classification: Planned | **Governing Source (added in the DS-005-A01 repair):** DS-PRT-002 (Transaction Recording, Planned) — requires immutable transaction recording that DS-PRT-001/003 derive positions and performance from; DS-ARC-023 (Portfolio Architecture, Planned) requires a Transaction/Position data model; DS-DB-023 (Committed, Append-Only Audit and Transaction Tables) already anticipates this entity.

Purpose: The authoritative, append-only transaction ledger Position (DS-DB-009) derives from, and the source of truth for realized/unrealized performance (DS-PRT-003).

Key Fields: transaction_id (primary identifier), portfolio_id (FK → Portfolio, DS-DB-008), symbol_id (FK → SecurityIdentity, DS-DB-014), transaction_type (buy/sell/dividend/split-adjustment/fee/other), quantity, price, fee (nullable), executed_at, recorded_at, source (manual/import — DS-DAT-002), reverses_transaction_id (nullable, self-referential FK → Transaction, for corrections).

Key Relationships: Many-to-one with Portfolio and SecurityIdentity; Position (DS-DB-009) is computed deterministically by aggregating a symbol's Transactions within a Portfolio; DS-PRT-003's realized/unrealized performance is likewise derived from this ledger, not stored independently.

Constraints/Invariants: Append-only (DS-DB-023) — a confirmed Transaction is never updated or deleted; corrections use an explicit `reverses_transaction_id`-linked adjusting entry (DS-PRT-002's "explicit, logged reversing/adjusting entry, never silent edits"). `quantity`/`price`/`fee` are the deterministic inputs DS-PRD-004 requires Position/performance calculations to reproduce exactly given the same transaction set.

Testing: Append-only/no-update-or-delete enforcement test; reversal-chain integrity test; position-derivation determinism test against fixture transaction sequences (shared with DS-DB-009's own test).

DS-DB-010 — Order

Release Classification: Planned | **Governing Source:** `ARCHITECTURE.md` §6, §14; `ROADMAP.md` Phase 7; DS-ARC-011

Purpose: Represent an order submitted to a Broker Adapter (paper initially) after passing the full `TradeValidationPipeline`.

Key Fields: order_id (unique, idempotency key per `TRADING_RULES.md` "Duplicate Order Protection"), trade_decision_id (FK), broker_environment (paper/live), symbol_id, quantity, order_type, status (pending/filled/partial/cancelled/rejected), submitted_at, filled_at (nullable), fill_price (nullable).

Key Relationships: One-to-one with the TradeDecision that produced it; many-to-one with BrokerState (DS-DB-012) for reconciliation.

Constraints/Invariants: `order_id` is globally unique and idempotent — a retry must not create a duplicate order (`TRADING_RULES.md`); `broker_environment = live` is disallowed until Stage 4/Phase 13 prerequisites are met (same gate as DS-DB-008).

Testing: Idempotency/duplicate-submission test (`ROADMAP.md` Phase 7 exit criterion).

DS-DB-011 — RiskState

Release Classification: Planned | **Governing Source:** `ARCHITECTURE.md` §6, §17; `ROADMAP.md` Phase 1 ("Risk-engine foundation"), Phase 7; DS-RSK-001/002

Purpose: Represent the current computed risk posture (per-trade, daily, weekly, portfolio-level) the Risk Engine evaluates against.

Key Fields: risk_state_id, portfolio_id (FK), as_of_timestamp, daily_loss_used, weekly_loss_used, open_position_count, sector_concentration (JSON), correlated_exposure

(JSON), drawdown_by_strategy (JSON).

Key Relationships: Read by the Risk Engine stage of the TradeValidationPipeline (DS-ARC-011, DS-ARC-012); one current RiskState per Portfolio, with historical snapshots retained for audit.

Constraints/Invariants: Computed deterministically (DS-RSK-002, now Committed per DS-002 v0.4.0); a stale RiskState (computed before the most recent Order) is not used to gate a new trade — the Risk Engine recomputes rather than trusting a cached snapshot for gating decisions.

Testing: Deterministic recomputation regression test (DS-RSK-002).

DS-DB-012 — BrokerState

Release Classification: Planned (corrected in the DS-005-A03 repair — `ROADMAP.md` Phase 7 paper-broker/reconciliation architecture is approved Planned work, not Future/Exploratory; live-mode use remains gated behind Phase 13) | **Governing Source:** `ARCHITECTURE.md` §6, §14; `ROADMAP.md` Phase 7 (paper), Phase 13 (live); DS-ARC-021 (Broker and Auto-Trader Architecture, Planned)

Purpose: Represent DarkSage's internal view of broker-side account state, for reconciliation (`TRADING_RULES.md`/`SECURITY_RULES.md` "Broker Reconciliation").

Key Fields: broker_state_id, portfolio_id (FK), broker_environment, positions_snapshot (JSON), cash_snapshot, open_orders_snapshot (JSON), last_reconciled_at, reconciliation_status (ok/mismatch).

Key Relationships: One-to-one (current) with Portfolio per environment; compared against Position/Order tables during reconciliation.

Constraints/Invariants: A `reconciliation_status = mismatch` pauses new trading and raises an audit event (`SECURITY_RULES.md` "Broker Reconciliation"), consistent with the fail-closed principle.

Testing: Not yet applicable — Paper Broker reconciliation is Phase 7; live is Phase 13. Recorded now because the entity shape is already implied by existing rules, not because it is committed.

9. Chart Entities

DS-DB-013 — ChartAnnotation

Release Classification: Planned | **Governing Source:** `ARCHITECTURE.md` §6, §9; DS-CHT-003

Purpose: Store chart annotations (trendlines, notes, markers), authored either by the user directly or by Sage, per DS-CHT-003.

Key Fields: annotation_id, symbol_id (FK), chart_config_id (nullable), type (trendline/note/marker/fibonacci), geometry (JSON — anchored to price/time, not pixel position), author (user/ai), created_at.

Key Relationships: Many-to-one with SecurityIdentity; independent of price/indicator data (presentation-layer only, per DS-PRD-003 — annotations never affect calculation).

Constraints/Invariants: `geometry` is anchored to price/time coordinates so it remains correctly positioned across timeframe/rescale changes (DS-CHT-003's edge case).

Testing: Rescale-anchoring regression test (DS-CHT-003).

10. Product and Platform Entities

DS-DB-014 — SecurityIdentity

Release Classification: Committed / MVP | **Governing Source:** DS-DAT-003 (Committed per DS-002 v0.4.0); `ARCHITECTURE.md` §6

Purpose: The canonical internal identity for a tradable security, decoupled from its display ticker, so a ticker/symbol change doesn't break references.

Key Fields: symbol_id (canonical, immutable, e.g., a surrogate key or FIGI-style identifier), current_ticker, instrument_type (stock/ETF/option/crypto/future), exchange, delisted_at (nullable).

Key Relationships: Referenced by nearly every other entity in this document (Candle, Quote, Signal, Position, Watchlist, ChartAnnotation) as the stable foreign key.

Constraints/Invariants: `symbol_id` is never reused after a delisting/reuse event, even if `current_ticker` is later reused by a different company (DS-DAT-003's disambiguation edge case).

Testing: Ticker-reuse disambiguation test (DS-DAT-003).

DS-DB-015 — UserProfile

Release Classification: Planned | **Governing Source:** DS-USR-006 (Planned per DS-002 v0.3.0)

Purpose: Minimal local user identity scoping preferences and workspace state, without requiring an external account.

Key Fields: profile_id, display_name, created_at.

Key Relationships: One-to-many with Preferences (DS-DB-016), WorkspaceLayout (DS-DB-017), Watchlist (DS-DB-018).

Constraints/Invariants: No external-account dependency for core local functionality (DS-USR-006).

Testing: Offline/no-account functional test.

DS-DB-016 — Preferences

Release Classification: Planned | **Governing Source:** DS-USR-002, DS-USR-003

Purpose: Persist user-configurable settings (terminology mode, notification settings, etc.).

Key Fields: profile_id (FK), key, value (JSON), updated_at.

Key Relationships: Many-to-one with UserProfile.

Constraints/Invariants: A corrupted/unreadable preference falls back to a documented safe default (DS-USR-002), not a crash.

Testing: Corrupted-storage recovery test.

DS-DB-017 — WorkspaceLayout

Release Classification: Planned (baseline shell is Committed per DS-WKS-001; saved multi-layout management remains Planned per DS-WKS-003) | **Governing Source:** DS-WKS-001, DS-WKS-003

Purpose: Persist workspace widget composition, position, and size.

Key Fields: layout_id, profile_id (FK), name, widget_composition (JSON), is_active, created_at.

Key Relationships: Many-to-one with UserProfile.

Constraints/Invariants: A layout referencing a widget type that no longer exists degrades gracefully on load (DS-WKS-003's edge case), not a load failure.

Testing: Missing-widget-type degradation test.

DS-DB-018 — Watchlist / WatchlistItem

Release Classification: Planned | **Governing Source:** DS-MKT-007 (Planned per DS-002 v0.4.0)

Purpose: A named, curated set of symbols a user tracks.

Key Fields: Watchlist: watchlist_id, profile_id (FK), name. WatchlistItem: watchlist_id (FK), symbol_id (FK), added_at.

Key Relationships: Many-to-many between UserProfile (via Watchlist) and SecurityIdentity.

Constraints/Invariants: A WatchlistItem referencing a delisted SecurityIdentity is flagged, not silently removed (DS-MKT-007).

Testing: Delisted-symbol flag test.

11. Security and Operations Entities

DS-DB-019 — IntegrationCredential

Release Classification: Committed / MVP | **Governing Source:** DS-INT-002 (Committed); DS-SEC-001; `SECURITY_RULES.md`

Purpose: Represent a stored external-integration credential (market-data provider, AI provider, future broker), using secure storage rather than plaintext.

Key Fields: credential_id, integration_type (market_data/ai_provider/broker), provider_name, secure_storage_reference (opaque pointer into OS credential store/encrypted vault — never the raw secret value itself), status (active/disabled/invalid), created_at, last_validated_at.

Key Relationships: Referenced by provider-adapter configuration (DS-ARC-006, DS-ARC-013).

Constraints/Invariants: The table never stores a raw secret value in a plain column — only an opaque reference to secure storage (DS-SEC-001, DS-ARC-015); no query against this table can return a usable secret.

Testing: Log/export scan test confirming absence of secret values (DS-SEC-001).

DS-DB-020 — AuditLogEntry

Release Classification: Committed / MVP | **Governing Source:** DS-OPS-001/002 (Committed)

Purpose: Append-only record of material application events (startup, ingestion failures, Risk Engine determinations, AI outages, crashes, security-sensitive actions).

Key Fields: entry_id, event_category, timestamp, correlated_entity_id (nullable, e.g., a TradeDecision or IntegrationCredential id), detail (JSON, secret-redacted), severity.

Key Relationships: Loosely referenced (by id, not FK constraint) from TradeDecision, RiskState, IntegrationCredential, and other event-producing entities.

Constraints/Invariants: Append-only — no update/delete path in normal operation; `detail` is redacted of secrets before write (DS-OPS-001, DS-SEC-001).

Testing: Log-content secret-redaction audit; audit-reconstruction test (DS-OPS-002).

12. Scanner Entities

DS-DB-021 — ScanConfiguration

Release Classification: Committed / MVP | **Governing Source:** DS-SCN-002 (Committed per DS-002 v0.4.0)

Purpose: A saved, reusable scan definition (universe, filter criteria, ranking configuration).

Key Fields: scan_id, profile_id (nullable FK — corrected in the DS-005-A03 repair; null resolves to the implicit single local profile when the Planned DS-USR-006 multi-profile feature is not yet enabled, so this Committed entity does not require a Planned feature to be independently satisfiable), name, universe_definition (JSON — e.g., watchlist reference, index membership, explicit list), filter_criteria (JSON), ranking_config (JSON), created_at.

Key Relationships: One-to-many with ScanResult (DS-DB-022).

Constraints/Invariants: `universe_definition` is evaluated against current membership at run time, with material changes disclosed (DS-SCN-002's edge case). A null `profile_id` is valid and represents scan configurations owned by the implicit default local profile.

Testing: Empty-universe test; scan save/execute regression test.

DS-DB-022 — ScanResult

Release Classification: Planned | **Governing Source:** DS-SCN-005 (Planned)

Purpose: Historical record of a scan execution's output, for trend review.

Key Fields: result_id, scan_id (FK), executed_at, matched_symbols (JSON, with per-symbol rank/score), universe_snapshot_size.

Key Relationships: Many-to-one with ScanConfiguration.

Constraints/Invariants: Immutable once written (historical results remain historical, DS-SCN-005); subject to a documented retention policy to bound storage growth.

Testing: Scan-history retrieval test confirming historical immutability.

13. Alerts and Notifications

Added in the DS-005-A02 repair to close a gap: DS-ALT-001/002 (Planned) require configured alerts and retrievable notification history/recovery, but no prior DS-005 entity supported them (Signal referenced "Alert" without a defined target).

DS-DB-026 — Alert

Release Classification: Planned | **Governing Source:** DS-ALT-001 (User-Configured Alerts, Planned)

Purpose: A user-configured, persistent alert condition (market data, watchlist event, or risk-limit condition per DS-ALT-001).

Key Fields: alert_id (primary identifier), profile_id (nullable FK → UserProfile, DS-DB-015 — same Phase-1-independence pattern as DS-DB-021 ScanConfiguration), name, condition_definition (JSON — e.g., symbol/indicator/threshold or risk-limit reference), status (active/disabled), created_at, last_triggered_at (nullable).

Key Relationships: One-to-many with Notification (DS-DB-027); optionally references a SecurityIdentity (DS-DB-014) or RiskState (DS-DB-011) condition target within `condition_definition`.

Constraints/Invariants: An Alert firing never creates an Order or any execution-path record — it produces only a Notification (DS-PRD-007's notification-only boundary; DS-EXE-001's pipeline-is-the-only-path-to-an-order boundary remains untouched by this entity).

Testing: Not yet applicable at Planned classification; deferred to Phase-appropriate authoring alongside DS-ALT-001's own implementation.

DS-DB-027 — Notification

Release Classification: Planned | **Governing Source:** DS-ALT-001 (fired-alert record), DS-ALT-002 (Notification Delivery, Planned)

Purpose: The durable record of a fired alert or system event, sufficient to survive an application restart so a notification missed while closed/backgrounded is visible on next open (DS-ALT-002's edge case).

Key Fields: notification_id (primary identifier), alert_id (nullable FK → Alert — nullable because a Notification may also originate from a non-Alert system event, e.g., a Risk Engine warning per DS-RSK-003 once that becomes Committed), triggered_at, condition_snapshot (JSON — the specific condition/value that triggered it, per DS-ALT-001's "timestamp, condition, triggering value"), delivered_at (nullable), read_at (nullable), channel (in_app / external — both Planned: DS-ALT-002 governs in-app delivery, DS-ALT-003 governs external-channel delivery).

Key Relationships: Many-to-one with Alert (when alert-originated); loosely referenced from other event-producing entities (RiskState, TradeDecision) the same way AuditLogEntry is (DS-DB-020), by id rather than a hard FK constraint.

Constraints/Invariants: Persisted immediately on trigger (not only on delivery) so that an application closed at fire time still shows the notification on next open, per DS-ALT-002's restart-recovery requirement; a Notification is never itself a trade/order record (same DS-PRD-007 boundary as Alert).

Testing: Not yet applicable at Planned classification; deferred to Phase-appropriate authoring. Restart-recovery test (notification visible after relaunch) is the key acceptance test once implemented.

14. Cross-Cutting Data Integrity Requirements

DS-DB-023 — Append-Only Audit and Transaction Tables

Priority: Critical | **Release Classification:** Committed / MVP | **Governing Source:** DS-PRD-004; DS-OPS-002; `AGENTS.md` "Database Changes"

Description: Tables recording historical fact (AuditLogEntry, BacktestResult, Transaction — DS-DB-025) shall be append-only in normal operation; corrections use an explicit, logged reversing/adjusting entry, never an in-place edit or delete.

Acceptance Criteria: No application code path issues an UPDATE or DELETE against these tables outside a documented, audited correction procedure.

Testing: Static analysis / code-review check for UPDATE/DELETE statements against append-only tables.

DS-DB-024 — Migration Discipline

Priority: High | **Release Classification:** Committed / MVP | **Governing Source:** `AGENTS.md` "Database Changes"

Description: Once migrations are introduced, all schema changes shall use versioned migrations; destructive schema changes shall require impact review and a documented migration/backup path.

Acceptance Criteria: No schema change ships without a corresponding migration script and rollback/backup note once the migration system exists.

Testing: Migration dry-run/rollback test (once the migration system exists; not yet applicable pre-Phase-1 tooling selection).

14a. Research, Journal, and Trade Intelligence Entities (added in the independent-audit H1 repair)

Closes a gap identified by independent audit: `TradeIntelligencePackage`, `ResearchSource`/`ResearchEvidence`, `CatalystEvent`, `TradingThesis`/`ThesisRevision`, `JournalEntry`, and `DailyReview`/`WeeklyReview` were named only in prose (§20) with no individually specified entity — this section gives each a full entity specification matching every other entity in this document.

DS-DB-028 — TradeIntelligencePackage

Release Classification: Planned | **Governing Source:** DS-SIG-005 (DS-002, Planned); DS-ARC-028 (DS-004, Planned)

Generated from DS-005 v0.5.0 — baseline 8b5cf5bd0341d5d6f70c67302c8bce2069d0929a

Purpose: Store the canonical Trade Intelligence Package DS-SIG-005 defines, as a single versioned object referenced identically by Sage, charts, journal, validation, execution, alerts, and audit.

Key Fields: package_id (primary identifier), version, signal_id (FK → Signal, DS-DB-004), symbol_id (FK), direction, strategy_id (FK, nullable), entry_zone (JSON: low/high or single value), stop, targets (JSON list), position_size, capital_at_risk, risk_reward, confidence, quality_rating, expected_holding_period, catalyst_ids (JSON list, FK → CatalystEvent, DS-DB-030), expires_at (nullable), evidence_ids (JSON list, FK → ResearchEvidence, DS-DB-029), contradictions (JSON), assumptions (JSON), invalidation_conditions (JSON), regime_id (FK → MarketRegime, DS-DB-003, nullable), data_freshness_state (enum, per DS-PRD-008), account_context (FK → Portfolio, DS-DB-008, nullable), approval_state, automation_mode (enum, per DS-EXE-008), generated_at.

Key Relationships: One-to-one with the originating Signal; many-to-one with StrategyProfile; referenced by JournalEntry (DS-DB-032), ChartAnnotation overlays (DS-DB-013), Alert/Notification (DS-DB-026/027), and TradeDecision (DS-DB-007) where a package proceeds to a proposal.

Constraints/Invariants: A field with no current value is explicitly null, never fabricated (DS-SIG-005); deterministic fields (capital_at_risk, risk_reward, position_size) are written only by the owning engine (Risk Engine, Portfolio) per DS-ARC-028, never by a generative process; package_id/version together are immutable once referenced by a downstream consumer — a material change creates a new version, not an in-place edit. **Security (DS-SCA-029, added for independent-audit Blocker 2):** each field additionally records its provenance (which service wrote it); the record is subject to the same tamper-detection discipline DS-SCA-027 applies to AuditLogEntry, extended to TradeIntelligencePackage versions; a package whose `data_freshness_state` has expired or whose assembly was partial is never treated as complete/current by a consuming query.

Testing: Cross-surface consistency test (shared with DS-ARC-028's own test); field-provenance audit; null-vs-fabricated-field regression test; tamper-detection test (shared with DS-SCA-029's own test).

DS-DB-029 — ResearchSource / ResearchEvidence

Release Classification: Planned | **Governing Source:** DS-RSH-001, DS-RSH-002 (DS-002, Planned); DS-ARC-026 (DS-004, Planned)

Purpose: Store ingested research items with full source/provenance/freshness metadata, per DS-RSH-001.

Key Fields: ResearchSource: source_id (primary identifier), provider_name, domain (enum: news/filing/earnings/macro/insider/political/analyst — DS-RSH-002), licensing_reference. ResearchEvidence: evidence_id (primary identifier), source_id (FK), affected_symbol_ids (JSON list, FK → SecurityIdentity), evidence_type, publication_time, event_time (nullable), retrieval_time, confidence, content_summary, raw_reference (opaque pointer to stored original), superseded_by (nullable, self-referential FK, for corrections).

Key Relationships: Many ResearchEvidence per ResearchSource; referenced by TradeIntelligencePackage (DS-DB-028), TradingThesis (DS-DB-031), and CatalystEvent (DS-DB-030).

Generated from DS-005 v0.5.0 — baseline 8b5cf5bd0341d5d6f70c67302c8bce2069d0929a

Constraints/Invariants: Immutable once stored except through an explicit, logged correction (`superseded_by`), per DS-ARC-026; conflicting evidence for the same claim is stored as separate records, never collapsed into one (DS-RSH-005); a domain that is disabled excludes its ResearchSource records from active queries without deleting historical ResearchEvidence.

Testing: Evidence-immutability audit (shared with DS-ARC-026's own test); conflicting-evidence disclosure test.

DS-DB-030 — CatalystEvent

Release Classification: Planned | **Governing Source:** DS-RSH-003 (DS-002, Planned)

Purpose: Represent a time-ordered, expected-or-realized market/company event exposed by the Catalyst and Event Timeline requirement (DS-RSH-003).

Key Fields: catalyst_id (primary identifier), affected_symbol_ids (JSON list), event_type, expected_date (nullable), realized_date (nullable), date_uncertainty (enum), source_evidence_ids (JSON list, FK → ResearchEvidence), post_event_update_ids (JSON list, self-referential).

Key Relationships: Referenced by TradeIntelligencePackage's `catalyst_ids` and by TradingThesis (DS-DB-031) as thesis-supporting evidence.

Constraints/Invariants: An event with an unresolved date discloses its uncertainty explicitly rather than presenting a guessed date as confirmed (DS-RSH-003).

Testing: Catalyst-timeline ordering and disclosure test.

DS-DB-031 — TradingThesis / ThesisRevision

Release Classification: Planned | **Governing Source:** DS-RSH-004 (DS-002, Planned); DS-ARC-026 (DS-004, Planned)

Purpose: Store a user- or Sage-assisted thesis with its supporting evidence, contradictions, assumptions, and invalidation conditions, preserving the original text as material evidence changes (DS-RSH-004).

Key Fields: TradingThesis: thesis_id (primary identifier), symbol_id (FK), original_text, assumptions (JSON), invalidation_conditions (JSON), evidence_ids (JSON list, FK → ResearchEvidence), created_at. ThesisRevision: revision_id (primary identifier), thesis_id (FK), revision_type (annotation/material-change-flag/status-update), content, triggering_evidence_id (nullable, FK → ResearchEvidence), created_at.

Key Relationships: One-to-many with ThesisRevision; referenced by JournalEntry (DS-DB-032) as the original plan's thesis link.

Constraints/Invariants: `original_text` is never overwritten; a material evidence change produces a new ThesisRevision, never a silent rewrite of the thesis (DS-RSH-004's "surface material changes without silently rewriting the original thesis").

Testing: Thesis-immutability and revision-chain integrity test (shared pattern with DS-DB-025's reversal-chain test).

DS-DB-032 — JournalEntry

Release Classification: Planned | **Governing Source:** DS-JRN-001, DS-JRN-002 (DS-002, Planned); DS-ARC-027 (DS-004, Planned)

Generated from DS-005 v0.5.0 — baseline 8b5cf5bd0341d5d6f70c67302c8bce2069d0929a

Purpose: Store a structured trade journal entry — original plan, actual execution, outcome, and lessons — with the original-plan immutability DS-JRN-002 requires.

Key Fields: entry_id (primary identifier), thesis_id (nullable, FK → TradingThesis), strategy_id (nullable, FK, version-pinned), original_plan (JSON: entry/stop/targets/size, written once), chart_state_reference (nullable), evidence_ids (JSON list), emotional_context (nullable, free text), actual_execution (JSON, populated after the fact), outcome (JSON, populated after close), rule_adherence (enum/JSON), mistake_classification (JSON list, from a defined vocabulary), lessons (text), created_at.

Key Relationships: One-to-one (optional) with TradingThesis; referenced by DailyReview/WeeklyReview (DS-DB-033) aggregation.

Constraints/Invariants: `original_plan` is written once and never overwritten; later edits, annotations, and Sage observations are separate, timestamped, attributable amendment records referencing this entry (DS-JRN-002) — append-only, mirroring DS-DB-023's pattern; an entry may represent a decision *not* to trade, under the same immutability discipline (DS-JRN-001). **Retention/deletion (DS-JRN-007, added for independent-audit Blocker 3):** a user-initiated deletion removes this record's private/emotional-context fields and its attachment references; it never deletes a Transaction (DS-DB-025) or AuditLogEntry (DS-DB-020) this entry references — those remain governed by their own append-only constraints (DS-DB-023) independent of journal deletion. A deletion under a documented legal/audit exception retains only the referenced immutable record's factual fields, never the JournalEntry's private narrative fields.

Testing: Original-plan immutability and amendment-attribution test (shared with DS-ARC-027's own test); deletion-propagation and immutable-record-preservation test (DS-JRN-007).

DS-DB-033 — DailyReview / WeeklyReview

Release Classification: Planned | **Governing Source:** DS-JRN-003, DS-JRN-004 (DS-002, Planned); DS-ARC-027 (DS-004, Planned)

Purpose: Store generated daily/weekly review aggregates over JournalEntry and deterministic performance data (DS-PERF), distinguishing deterministic figures from Sage's advisory commentary.

Key Fields: DailyReview: review_id (primary identifier), review_date, performance_metric_refs (JSON, FK references into DS-PERF's own metrics — never independently computed here), plan_adherence_summary (JSON), recurring_mistake_refs (JSON list), next_session_watchlist (JSON list, FK → SecurityIdentity), sage_commentary (nullable text, clearly separated from performance_metric_refs), generated_at. WeeklyReview: analogous fields aggregated over a week, plus `pattern_discovery_notes` (text, explicitly non-causal per DS-JRN-004).

Key Relationships: Many-to-one aggregation over JournalEntry (DS-DB-032) and DS-PERF's performance metrics for the same date range.

Constraints/Invariants: Every quantitative figure traces to a DS-PERF/DS-PRT deterministic calculation by reference, never recomputed or restated independently by this entity or by Sage (DS-PRD-004); `sage_commentary/pattern_discovery_notes` are structurally distinct fields from the deterministic references, never merged into one narrative that obscures which is which. **Retention/deletion (DS-JRN-007):** deleting a source JournalEntry (DS-DB-032) propagates to any DailyReview/WeeklyReview narrative content (`sage_commentary/pattern_discovery_notes`) that would reproduce that entry's private text; the review's `performance_metric_refs` (deterministic, sourced from DS-PERF/DS-PRT independent of any single journal entry) are unaffected.

Testing: Deterministic-figure-reuse audit (shared with DS-ARC-027's own test); correlation-vs-causation content review for `pattern_discovery_notes` (DS-JRN-004).

15. Non-Goals

DS-005 does not: select a database engine beyond SQLite-first (DS-ARC-016 already governs this); define exact SQL column types/indexes (implementation detail); or define API serialization shapes (DS-006).

16. Dependencies

- DS-001, DS-002, DS-004
- `ARCHITECTURE.md` §6, §20; `PROJECT_SPEC.md` §7, §16, §17; `TRADING_RULES.md`; `SECURITY_RULES.md`; `AGENTS.md`

17. Risks and Constraints

- **Sequencing risk:** authored before DS-006 (API contracts); entity `_snapshot`/JSON-blob fields are provisional shape, refined once API serialization needs are known.
- **Classification discipline:** the DS-005-A03/A05 repair applied the same "Phase-1 core vs. later-phase extension" field-splitting pattern used for `DS-ARC-005`/`DS-ARC-004` (DS-004) and `DS-AI`/`DS-SGE` (DS-002/DS-003) to `StrategyProfile` and `TradeDecision`, so a Committed base model never requires a Planned capability to be valid.

18. Verification Approach

Entity-level Testing per DS-DB-NNN item. Document-level verification (unique-ID check, cross-reference consistency, no contradiction with DS-002/DS-003/DS-004) recorded in `.ai-workflow/HANDOFF.md`.

19. References

- `ARCHITECTURE.md`, `PROJECT_SPEC.md`, `TRADING_RULES.md`, `SECURITY_RULES.md`, `AGENTS.md`
- `docs/codex/Volume-02-Product/DS-002-SRS.md` and `requirements/*.md`
- `docs/codex/Volume-04-Architecture/DS-004-Technical-Architecture.md`

Appendix A — Open Questions

1. **RESOLVED in the DS-005-A01 repair.** Transaction entity (DS-DB-025) added; Position and DS-DB-023 updated to reference it. Retained here for revision-history traceability, not as an open item.
2. **JSON-blob field granularity** — several entities use JSON columns (e.g., `pipeline_stage_results`, `configuration`) as a placeholder for structured sub-schemas better suited to normalized tables. Acceptable for a first draft; a later revision should normalize

where query patterns justify it (DS-006/DS-004 input).

3. **RESOLVED in the DS-005-A02 repair.** Alert (DS-DB-026) and Notification (DS-DB-027) entities added, supporting DS-ALT-001/002 and preserving the DS-PRD-007 notification-only boundary. Retained here for revision-history traceability, not as an open item.
4. **Nullable `profile_id` pattern (DS-005-A03 repair)** — `ScanConfiguration` and `Alert` both use a nullable `profile_id` resolving to an implicit default local profile so they remain independently satisfiable without the Planned multi-profile `UserProfile` feature (DS-USR-006). This is a routine implementation pattern, not a governance question, but is noted here since it's a repeated design choice worth keeping consistent if more Committed entities gain a similar dependency in the future.

20. Founder Vision Completion Data Domains

The logical data model shall provide versioned entities for `TradeIntelligencePackage`, `ResearchSource`, `ResearchEvidence`, `CatalystEvent`, `TradingThesis`, `ThesisRevision`, `JournalEntry`, `DailyReview`, `WeeklyReview`, `SageTaskPlan`, `SageToolInvocation`, `MonitoringPolicy`, `NotificationChannel`, `NotificationDelivery`, and `DiscordChannelConfiguration`.

Original trade plans and thesis versions remain recoverable. Deterministic values record calculation version and inputs. Discord credentials are never stored in plaintext application records; configuration stores only protected secret references and non-secret routing metadata. Notification deliveries record idempotency key, event reference, channel, attempt state, timestamps, redacted response metadata, and final disposition.